

Maximize rho for dominating feasible self-maximizing computable shares under additive valuations

ChatGPT (gpt-5.6-sol)*

Solution generated on 2026-08-24

Abstract

We give a complete solution to the problem of determining the largest maximin-share domination factor achievable, as a supremum, by a feasible, self-maximizing, polynomial-time computable and polynomial-time allocable share for a fixed number $n \geq 3$ of agents.

Let

$$\alpha_n = \inf_{\substack{M \text{ finite, } (v_1, \dots, v_n) \\ \text{MMS}_n(v_i) > 0 \ \forall i}} \max_{(A_1, \dots, A_n) \in \mathcal{P}_n(M)} \min_{i \in [n]} \frac{v_i(A_i)}{\text{MMS}_n(v_i)}.$$

We prove that the required supremum is exactly

$$\rho_n^* = \alpha_n.$$

The upper bound follows directly from feasibility. For the converse, for every $\rho < \alpha_n$ we construct a finite universal collection of max-min menus. A bounded-mass normalization, geometric rounding, histogram compression, and lattice rounding reduce the menus to polynomially many permutation orbits. Their cross-feasibility follows from the definition of α_n , while a max-min menu lemma gives feasibility and self-maximization. We also give exact polynomial-time algorithms, for fixed n and ρ , for evaluating the share and producing an allocation meeting it.

Finally, we characterize α_n by a finite optimization over n^n atomic good types. This proves that α_n is an attained rational constant and is exactly computable in principle for every fixed n . The solution concerns the supremum requested in the problem; it does not assert that a single admissible share attains the endpoint α_n .

1 The open problem

This paper addresses an open problem posed by Moshe Babaioff, Uriel Feige in *Fair Shares: Feasibility, Domination and Incentives* [1] (Economics and Computation (EC), 2022), Page 7, Section 1.2 (Our Results), last paragraph; reiterated Page 26, Section 7 (Discussion), last paragraph.. The website catalogue entry is problem 23 (W4285090266_p0) of the [Open Problems in Operations Research](#) collection.

*Generated by the `base_solver_ni_two_iterations` pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve $\rightarrow$ A.8 verify $\rightarrow$ A.9 write-up; run `run_20260824_073025_b05799dd, ec-2it-ultra`); two-iteration run, outcome from iteration 2. Independent verifier assessment: APPROVED with progress score 3/3 (full solution). Maintained by the AI in Mathematics & Operations Research project.

For additive valuations over indivisible goods and a fixed number of agents $n \geq 3$, determine the supremum value $\rho^* \in [0, 1]$ such that there exists a share function s satisfying all three properties:

1) *Feasibility*: for every additive valuation profile $(v_1, \dots, v_n)$ there exists an allocation A with $v_i(A_i) \geq s(v_i, n)$ for all i .

2) *Self-maximizing*: for every additive true valuation v and additive report v' , $\hat{s}_v(v) \geq \hat{s}_v(v')$.

3) *Polynomial-time computability (and allocability)*: $s(v, n)$ can be computed in polynomial time from additive item values, and there is a polynomial-time algorithm that, on every additive valuation profile, computes an allocation A with $v_i(A_i) \geq s(v_i, n)$ for all i .

subject to the quantitative domination requirement

$$s(v, n) \geq \rho^* \text{MMS}_n(v) \quad \text{for all additive } v.$$

Equivalently, find the largest achievable domination factor ρ for which such a feasible, self-maximizing, polynomial-time computable share exists for additive valuations.

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

2 Framework and statement of the result

Fix an integer $n \geq 3$. Write $[n] = \{1, \dots, n\}$. Let M be a finite set of indivisible goods, with $m = |M|$. An additive valuation on M is a function $v : 2^M \rightarrow \mathbb{R}_{\geq 0}$ of the form

$$v(S) = \sum_{j \in S} v_j, \quad v_j \geq 0.$$

In particular, all valuations considered here are normalized by $v(\emptyset) = 0$ and are monotone.

Let $\mathcal{P}_n(M)$ denote the set of ordered partitions

$$(P_1, \dots, P_n)$$

of M , where empty parts are permitted. The n -maximin share of v is

$$\text{MMS}_n(v) = \max_{(P_1, \dots, P_n) \in \mathcal{P}_n(M)} \min_{k \in [n]} v(P_k).$$

Because M is finite, the maximum is attained.

An allocation is an ordered partition $(A_1, \dots, A_n) \in \mathcal{P}_n(M)$. We shall also temporarily use pairwise-disjoint partial allocations. Since valuations are monotone, every such partial allocation may be completed by assigning the remaining goods arbitrarily, without decreasing any agent's value.

Definition 1 (Shares and implied guarantees). For fixed n , a share rule assigns to every finite ground set M and every additive valuation v on M a number $s_M(v)$ satisfying

$$0 \leq s_M(v) \leq v(M).$$

Given a true valuation v and a reported valuation w , define the acceptable-set family

$$\mathcal{A}_s(w) = \{S \subseteq M : w(S) \geq s_M(w)\}$$

and the true guarantee implied by the report w as

$$\hat{s}_v(w) = \min_{S \in \mathcal{A}_s(w)} v(S).$$

The family $\mathcal{A}_s(w)$ is nonempty because $M \in \mathcal{A}_s(w)$.

Definition 2 (Feasibility and self-maximization). A share rule s is feasible if, for every profile $(v_1, \dots, v_n)$ on a common ground set M , there is an allocation $(A_1, \dots, A_n) \in \mathcal{P}_n(M)$ such that

$$v_i(A_i) \geq s_M(v_i) \quad \text{for every } i \in [n].$$

It is self-maximizing if truthful reporting maximizes the implied guarantee:

$$\hat{s}_v(v) \geq \hat{s}_v(w)$$

for every pair of additive valuations v, w on the same ground set.

For the computational assertions, valuations are represented by nonnegative rational item values in binary. A share is polynomial-time computable if $s_M(v)$ can be evaluated exactly in time polynomial in the total input length. It is polynomial-time allocable if, given a rational profile, an allocation meeting all the shares can be found in polynomial time. Throughout, n and the desired approximation factor are fixed constants; the degree and constant factors of the running-time bounds may depend on them.

Define

$$\Phi(v_1, \dots, v_n) = \max_{(A_1, \dots, A_n) \in \mathcal{P}_n(M)} \min_{i \in [n]} \frac{v_i(A_i)}{\text{MMS}_n(v_i)}$$

whenever all the maximin shares in the denominator are positive, and set

$$\alpha_n = \inf_{\substack{M \text{ finite, } (v_1, \dots, v_n) \\ \text{MMS}_n(v_i) > 0 \ \forall i}} \Phi(v_1, \dots, v_n). \quad (1)$$

For every profile occurring in the infimum, finiteness of the allocation set implies that the maximum defining Φ is attained. Moreover, $\Phi(v_1, \dots, v_n) \geq \alpha_n$. Thus every all-positive-MMS profile has an allocation satisfying

$$v_i(A_i) \geq \alpha_n \text{MMS}_n(v_i) \quad (i \in [n]). \quad (2)$$

Profiles containing zero-MMS agents satisfy the same conclusion: if at least one agent has positive MMS, duplicate one of the positive-MMS valuations until there are n positive-MMS valuations, apply (2), and retain the bundles corresponding to the original positive-MMS agents.

The zero-MMS agents impose no positive lower bound. If every agent has zero MMS, the assertion is vacuous.

We shall use the elementary bound

$$0 \leq \alpha_n \leq 1. \quad (3)$$

The lower bound is immediate. For the upper bound, take n goods, each of unit value, and give every agent the same valuation. Its maximin share is 1, and the largest possible minimum value in an allocation is also 1. Consequently, the corresponding value of Φ is 1.

Let ρ_n^* be the supremum of all $\rho \geq 0$ for which there exists a feasible, self-maximizing, polynomial-time computable and polynomial-time allocable share s satisfying

$$s_M(v) \geq \rho \text{ MMS}_n(v) \quad (4)$$

for every finite M and every additive valuation v on M .

Theorem 1 (Complete solution). *For every fixed integer $n \geq 3$,*

$$\boxed{\rho_n^* = \alpha_n.} \quad (5)$$

More precisely:

1. *no feasible share can satisfy (4) with $\rho > \alpha_n$;*
2. *for every real $\rho < \alpha_n$, there is a feasible, self-maximizing, polynomial-time computable and polynomial-time allocable share satisfying (4);*
3. *α_n is the value of a finite optimization depending only on n , is rational, and is exactly computable in principle.*

Equality (5) is an equality of suprema. No assertion of endpoint attainment by a single admissible share is made or needed.

3 The upper bound

Proposition 1. *If a feasible share s satisfies*

$$s_M(v) \geq \rho \text{ MMS}_n(v)$$

for every additive valuation v , then $\rho \leq \alpha_n$.

Proof. Take any profile $(v_1, \dots, v_n)$ for which all maximin shares are positive. Feasibility supplies an allocation $(A_1, \dots, A_n)$ such that

$$v_i(A_i) \geq s_M(v_i) \geq \rho \text{ MMS}_n(v_i) \quad (i \in [n]).$$

Therefore

$$\Phi(v_1, \dots, v_n) \geq \rho.$$

Taking the infimum over all such profiles gives $\alpha_n \geq \rho$. □

It follows immediately that

$$\rho_n^* \leq \alpha_n. \quad (6)$$

4 A max–min menu lemma

The lower bound will be obtained from a finite collection of universal menus.

Lemma 1 (Max–min menu lemma). *Let $\mathcal{F}$ be a finite nonempty collection whose members are nonempty families $F \subseteq 2^M$. Assume the following cross-feasibility property: for every ordered n -tuple*

$$F_1, \dots, F_n \in \mathcal{F},$$

with repetitions permitted, there are pairwise-disjoint sets $S_1, \dots, S_n$ such that

$$S_i \in F_i \quad (i \in [n]). \quad (7)$$

Define

$$s_{\mathcal{F}}(v) = \max_{F \in \mathcal{F}} \min_{S \in F} v(S). \quad (8)$$

Then $s_{\mathcal{F}}$ is feasible and self-maximizing.

Proof. For a report w , choose a family $F_w \in \mathcal{F}$ attaining the maximum in (8). Every $S \in F_w$ satisfies

$$w(S) \geq \min_{T \in F_w} w(T) = s_{\mathcal{F}}(w).$$

Consequently,

$$F_w \subseteq \{S \subseteq M : w(S) \geq s_{\mathcal{F}}(w)\}.$$

For any true valuation v , it follows that

$$\widehat{s}_v(w) \leq \min_{S \in F_w} v(S) \leq \max_{F \in \mathcal{F}} \min_{S \in F} v(S) = s_{\mathcal{F}}(v). \quad (9)$$

Under truthful reporting, every acceptable set has true value at least $s_{\mathcal{F}}(v)$, and therefore

$$\widehat{s}_v(v) \geq s_{\mathcal{F}}(v). \quad (10)$$

Equations (9) and (10) prove self-maximization.

For feasibility, given a profile $(v_1, \dots, v_n)$, choose for every i a maximizing family $F_i \in \mathcal{F}$. By (7), there are pairwise-disjoint sets $S_i \in F_i$. Then

$$v_i(S_i) \geq \min_{S \in F_i} v_i(S) = s_{\mathcal{F}}(v_i).$$

Assign the unallocated goods arbitrarily. Monotonicity preserves all these inequalities, so the resulting complete allocation meets every share. $\square$

5 Choice of constants

Fix a real number $\rho < \alpha_n$. If ρ is irrational, choose a rational number ρ' satisfying

$$\rho < \rho' < \alpha_n.$$

A share dominating ρ' MMS $_n$ also dominates ρ MMS $_n$. It therefore suffices to construct the share when ρ is rational, and we assume this from now on.

Choose a rational number a such that

$$\rho < a < \alpha_n. \quad (11)$$

By (3), $0 < a < 1$. Choose rational constants

$$c > 1, \quad \varepsilon > 0, \quad \gamma > 0, \quad 0 < \eta < 1$$

small enough that, with

$$b = 1 - \varepsilon - \gamma, \quad (12)$$

we have

$$b > 0 \quad \text{and} \quad ab - \varepsilon - \eta > c\rho. \quad (13)$$

Such rational constants exist. Indeed, as

$$c \downarrow 1, \quad \varepsilon, \gamma, \eta \downarrow 0,$$

the left side of the second inequality in (13) tends to a , while its right side tends to ρ , and $a > \rho$.

All constants from this point onward depend only on the fixed n and ρ , and not on m or on the reported valuations.

6 Bounded-mass normalization

Lemma 2 (Capped MMS normalization). *Let v be an additive valuation with*

$$\mu = \text{MMS}_n(v) > 0.$$

Define

$$y_j = \min \left\{ \frac{v_j}{\mu}, 1 \right\} \quad (j \in M). \quad (14)$$

Then

$$\text{MMS}_n(y) = 1. \quad (15)$$

Proof. Choose a partition $(P_1, \dots, P_n) \in \mathcal{P}_n(M)$ attaining $\text{MMS}_n(v) = \mu$. For every $k \in [n]$,

$$v(P_k) \geq \mu.$$

If P_k contains an item j with $v_j \geq \mu$, then $y_j = 1$, so $y(P_k) \geq 1$. If it contains no such item, capping changes none of its normalized item values and

$$y(P_k) = \frac{v(P_k)}{\mu} \geq 1.$$

Thus this partition shows that $\text{MMS}_n(y) \geq 1$.

On the other hand,

$$y_j \leq \frac{v_j}{\mu} \quad (j \in M).$$

Monotonicity of the maximin share under coordinatewise domination gives

$$\text{MMS}_n(y) \leq \text{MMS}_n(v/\mu) = \frac{\text{MMS}_n(v)}{\mu} = 1.$$

Combining the two inequalities proves (15). □

Lemma 3 (Uniform total-mass bound). *If $0 \leq y_j \leq 1$ for all j and $\text{MMS}_n(y) = 1$, then*

$$\sum_{j \in M} y_j \leq 2n - 1. \quad (16)$$

Proof. Assign the items successively to n bins, always assigning the next item to a currently least-loaded bin. Let $L_{\min}$ be the smallest final load.

Every final bin load is at most $L_{\min} + 1$. To see this, consider the last item placed in a given nonempty bin. Immediately before that item was placed, the bin was least-loaded. Its load at that moment was therefore no larger than the then-current load of a bin that eventually has final load $L_{\min}$, and hence no larger than $L_{\min}$. The last assigned item has value at most 1. Empty bins also satisfy the asserted bound.

Writing

$$T = \sum_{j \in M} y_j,$$

one bin has load $L_{\min}$, and each of the other $n - 1$ bins has load at most $L_{\min} + 1$. Hence

$$T \leq L_{\min} + (n - 1)(L_{\min} + 1) = nL_{\min} + n - 1,$$

so

$$L_{\min} \geq \frac{T - (n - 1)}{n}. \quad (17)$$

If $T > 2n - 1$, then (17) gives $L_{\min} > 1$. The greedy assignment would then be a partition all of whose bundles have value greater than 1, contradicting $\text{MMS}_n(y) = 1$. Thus (16) holds. $\square$

The non-strict inequality in (16) is necessary: for example, $2n - 1$ unit items have total mass $2n - 1$ and maximin share 1.

7 Geometric rounding and histogram compression

We now construct, for every ground-set size $m \geq 1$, a polynomially sized collection of weight multisets.

Set

$$\delta = \frac{\eta}{m} \quad (18)$$

and let L be the least positive integer such that

$$c^L \geq \frac{m}{\eta}. \quad (19)$$

Equivalently,

$$L = \left\lceil \log_c \frac{m}{\eta} \right\rceil.$$

Because L is minimal,

$$\delta < c^{-(L-1)} \leq c\delta. \quad (20)$$

Define the set of levels

$$W_m = \{\delta, c^{-(L-1)}, c^{-(L-2)}, \dots, c^{-1}, 1\}, \quad (21)$$

with any repetitions removed, and write its distinct elements in increasing order as

$$0 < w_1 < \cdots < w_d \leq 1.$$

Since c and η are fixed,

$$d = O_{n,\rho}(\log(m+1)). \quad (22)$$

Given the normalized vector y from (14), round each coordinate upward to the least level in W_m that is at least y_j . Denote the resulting vector by z . For $y_j = 0$, this rule gives $z_j = \delta$. For $0 < y_j \leq \delta$, it also gives $z_j = \delta$. For $y_j > \delta$, adjacent geometric levels differ by a factor c , with (20) covering the transition from δ to the first geometric level. Consequently, for every item,

$$y_j \leq z_j \leq cy_j + \delta. \quad (23)$$

It follows from $z \geq y$ and Lemma 5.1 that

$$\text{MMS}_n(z) \geq 1. \quad (24)$$

Using (16), (18), and (23), we also obtain

$$\sum_{j \in M} z_j \leq c \sum_{j \in M} y_j + m\delta \leq c(2n-1) + \eta.$$

Set

$$B = c(2n-1) + \eta. \quad (25)$$

A sorted W_m -valued vector is described by counts

$$N_\ell = |\{j : z_j = w_\ell\}|, \quad \ell \in [d],$$

and by its mass histogram

$$p_\ell = w_\ell N_\ell. \quad (26)$$

Equation (25) says

$$\sum_{\ell=1}^d p_\ell \leq B. \quad (27)$$

Put

$$\theta = \frac{\varepsilon}{2}, \quad \lambda = \frac{\theta}{d}, \quad R = \left\lceil \frac{B}{\lambda} \right\rceil, \quad (28)$$

and define the finite simplex grid

$$G_m = \left\{ \lambda k : k \in \mathbb{Z}_{\geq 0}^d, \sum_{\ell=1}^d k_\ell \leq R \right\}. \quad (29)$$

Because B and θ are fixed and $\lambda = \theta/d$, we have $R = O_{n,\rho}(d)$. The number of grid points is therefore

$$|G_m| = \binom{R+d}{d} = \exp(O_{n,\rho}(d)) = m^{O_{n,\rho}(1)}. \quad (30)$$

For every $g = (g_1, \dots, g_d) \in G_m$, choose, with deterministic tie-breaking, nonnegative integer counts

$$q_1(g), \dots, q_d(g), \quad \sum_{\ell=1}^d q_\ell(g) \leq m,$$

that minimize

$$\sum_{\ell=1}^d |w_\ell q_\ell(g) - g_\ell|. \quad (31)$$

The expanded representative consists of $q_\ell(g)$ copies of w_ℓ , padded with zeroes to obtain a vector $r^0(g)$ of length m . We always write this vector in nonincreasing order.

Lemma 4 (Histogram approximation). *For every W_m -valued target vector z satisfying (25), some $g \in G_m$ has a representative $r^0(g)$ such that, after both vectors are sorted in the same order,*

$$\|z - r^0(g)\|_1 < \varepsilon. \quad (32)$$

Proof. Let p be the target mass histogram (26), and define

$$g_\ell = \lambda \left\lfloor \frac{p_\ell}{\lambda} \right\rfloor. \quad (33)$$

Then

$$\sum_{\ell=1}^d \frac{g_\ell}{\lambda} = \sum_{\ell=1}^d \left\lfloor \frac{p_\ell}{\lambda} \right\rfloor \leq \frac{1}{\lambda} \sum_{\ell=1}^d p_\ell \leq \frac{B}{\lambda} \leq R.$$

Thus $g \in G_m$, and

$$\|p - g\|_1 = \sum_{\ell=1}^d (p_\ell - g_\ell) < d\lambda = \theta. \quad (34)$$

The target counts N_ℓ are feasible in the minimization (31), since $\sum_\ell N_\ell = m$. If

$$p'_\ell = w_\ell q_\ell(g)$$

is the histogram selected by (31), optimality gives

$$\|p' - g\|_1 \leq \|p - g\|_1 < \theta.$$

Hence

$$\|p - p'\|_1 \leq \|p - g\|_1 + \|g - p'\|_1 < 2\theta = \varepsilon. \quad (35)$$

It remains to pass from histogram distance to coordinate distance. Match as many equal-level items as possible. At level w_ℓ , the total mass left unmatched is

$$w_\ell |N_\ell - q_\ell(g)|.$$

After zero-padding, the remaining coordinates on both sides can be matched bijectively. For every two nonnegative numbers x, y ,

$$|x - y| \leq x + y,$$

so the cost of any such matching is at most the total unmatched mass, namely $\|p - p'\|_1$. Matching two sorted real sequences minimizes their one-dimensional ℓ_1 matching cost: an inversion can be removed using

$$|x - a| + |y - b| \leq |x - b| + |y - a| \quad (x \geq y, a \geq b).$$

Therefore the sorted coordinate matching has cost at most $\|p - p'\|_1$. Equation (35) proves (32). $\square$

Set

$$h = \frac{\gamma}{m}. \quad (36)$$

For every representative $r^0(g)$, round each coordinate downward to the lattice $h\mathbb{Z}_{\geq 0}$:

$$\bar{r}_j(g) = h \left\lfloor \frac{r_j^0(g)}{h} \right\rfloor. \quad (37)$$

Then

$$0 \leq \bar{r}_j(g) \leq r_j^0(g)$$

and

$$\|r^0(g) - \bar{r}(g)\|_1 < mh = \gamma. \quad (38)$$

Retain $\bar{r}(g)$ if and only if there is a partition

$$(P_1, \dots, P_n) \in \mathcal{P}_n(M)$$

such that

$$\bar{r}(g)(P_k) \geq b \quad (k \in [n]). \quad (39)$$

Let $\mathcal{R}_m$ be the set of distinct retained weight multisets.

For a retained multiset $\bar{r}$ and any assignment π of its m coordinates to the goods in M , write $\bar{r}^\pi$ for the resulting weight vector and define

$$F_{\bar{r}, \pi} = \{S \subseteq M : \bar{r}^\pi(S) \geq ab\}. \quad (40)$$

Also define

$$F_0 = \{\emptyset\}. \quad (41)$$

Since $a < 1$, every partition satisfying (39) has

$$\bar{r}(P_k) \geq b > ab.$$

Thus every family $F_{\bar{r}, \pi}$ is nonempty. Define the complete menu collection

$$\mathcal{F}_m = \{F_0\} \cup \{F_{\bar{r}, \pi} : \bar{r} \in \mathcal{R}_m, \pi \text{ an assignment to } M\}. \quad (42)$$

8 Cross-feasibility of the menu collection

Proposition 2. *For every $m \geq 1$, the collection $\mathcal{F}_m$ satisfies the cross-feasibility hypothesis of the max-min menu lemma.*

Proof. First consider an ordered n -tuple containing no copies of F_0 :

$$F_{\bar{r}_1, \pi_1}, \dots, F_{\bar{r}_n, \pi_n}.$$

For each i , retention gives

$$\text{MMS}_n(\bar{r}_i^{\pi_i}) \geq b. \quad (43)$$

Define an additive valuation

$$x_i = \frac{\bar{r}_i^{\pi_i}}{b}. \quad (44)$$

Then

$$\text{MMS}_n(x_i) \geq 1 > 0. \quad (45)$$

By the definition (1) of α_n , the profile $(x_1, \dots, x_n)$ has an allocation $(A_1, \dots, A_n)$ for which

$$\min_{i \in [n]} \frac{x_i(A_i)}{\text{MMS}_n(x_i)} \geq \alpha_n > a.$$

In particular, for every i ,

$$x_i(A_i) \geq a \text{ MMS}_n(x_i) \geq a.$$

Multiplying by b gives

$$\bar{r}_i^{\pi_i}(A_i) \geq ab.$$

Thus

$$A_i \in F_{\bar{r}_i, \pi_i},$$

and the A_i are pairwise disjoint.

Now suppose the tuple contains exactly $k < n$ nonzero menus. If $k = 0$, use $\emptyset$ as the representative for every menu. If $1 \leq k < n$, form an n -agent profile by taking the k valuations (44) associated with the nonzero menus and duplicating any one of them $n - k$ times. Apply the preceding argument to this n -agent profile and retain the bundles corresponding to the original k valuations. Use $\emptyset$ for every copy of F_0 . These representatives are pairwise disjoint and belong to their respective menu families. $\square$

9 Existence of a dominating witness menu

Proposition 3 (Witness menu). *Let v be an additive valuation on a nonempty ground set M , and suppose*

$$\mu = \text{MMS}_n(v) > 0.$$

Then $\mathcal{F}_m$ contains a menu F satisfying

$$\min_{S \in F} v(S) \geq \rho \text{ MMS}_n(v). \quad (46)$$

In fact every $S \in F$ satisfies the strict inequality

$$v(S) > \rho \text{ MMS}_n(v). \quad (47)$$

Proof. Define y by (14), and construct its upward rounding z by (21)–(23). Order the goods so that

$$v_1 \geq v_2 \geq \cdots \geq v_m. \quad (48)$$

The coordinatewise maps

$$v_j \mapsto y_j \mapsto z_j$$

are nondecreasing, so y and z are also nonincreasing in this order.

Apply Lemma 6.1 to z . It provides a grid representative r^0 , sorted in the same order, such that

$$\|z - r^0\|_1 < \varepsilon. \quad (49)$$

Let $\bar{r}$ be its lattice rounding (37). By (38),

$$\|z - \bar{r}\|_1 \leq \|z - r^0\|_1 + \|r^0 - \bar{r}\|_1 < \varepsilon + \gamma. \quad (50)$$

By (24), there is a partition $(P_1, \dots, P_n)$ satisfying

$$z(P_k) \geq 1 \quad (k \in [n]).$$

For every k ,

$$\begin{aligned} \bar{r}(P_k) &\geq z(P_k) - \sum_{j \in P_k} |z_j - \bar{r}_j| \\ &\geq z(P_k) - \|z - \bar{r}\|_1 \\ &> 1 - \varepsilon - \gamma \\ &= b. \end{aligned} \quad (51)$$

Thus $\bar{r}$ is retained.

Assign the coordinates of $\bar{r}$ to the goods in the common order (48). The corresponding menu

$$F = F_{\bar{r}, \pi}$$

belongs to $\mathcal{F}_m$. Let $S \in F$. Then

$$\bar{r}^\pi(S) \geq ab. \quad (52)$$

Because $\bar{r} \leq r^0$ coordinatewise and (49) holds,

$$\begin{aligned} z(S) &\geq r^0(S) - \sum_{j \in S} |z_j - r_j^0| \\ &\geq \bar{r}^\pi(S) - \|z - r^0\|_1 \\ &> ab - \varepsilon. \end{aligned} \quad (53)$$

On the other hand, (23) and $|S| \leq m$ imply

$$z(S) \leq cy(S) + |S|\delta \leq cy(S) + m\delta = cy(S) + \eta. \quad (54)$$

Combining (53) and (54),

$$y(S) > \frac{ab - \varepsilon - \eta}{c}.$$

The choice (13) of constants gives

$$y(S) > \rho. \quad (55)$$

Finally, $y_j \leq v_j/\mu$ for every j , and therefore

$$\frac{v(S)}{\mu} \geq y(S) > \rho.$$

Since $\mu = \text{MMS}_n(v)$, this is precisely (47), and (46) follows. $\square$

For $m \geq 1$, define

$$s_{\rho, M}(v) = \max_{F \in \mathcal{F}_m} \min_{S \in F} v(S). \quad (56)$$

For the empty ground set, define

$$s_{\rho, \emptyset}(v) = 0. \quad (57)$$

Theorem 2 (Existence below α_n). *For every $\rho < \alpha_n$, the share s_ρ defined above is feasible and self-maximizing and satisfies*

$$s_{\rho, M}(v) \geq \rho \text{MMS}_n(v) \quad (58)$$

for every finite M and every additive valuation v .

Proof. For $M = \emptyset$, all quantities in (58) are zero, the empty allocation is feasible, and truthful reporting is trivially self-maximizing.

Suppose $m \geq 1$. Proposition 7.1 and the max–min menu lemma imply that (56) is feasible and self-maximizing. If $\text{MMS}_n(v) > 0$, Proposition 8.1 provides a menu whose floor is at least $\rho \text{MMS}_n(v)$, and taking the maximum in (56) proves (58). If $\text{MMS}_n(v) = 0$, then the right side of (58) is zero, while every menu floor and hence $s_{\rho, M}(v)$ is nonnegative. Thus (58) also holds in this case. $\square$

10 Polynomial-time computation and allocation

We next prove the required algorithmic properties. All arithmetic in this section is exact rational arithmetic.

10.1 Enumeration and retention

The integer L in (19) is found by repeated exact comparison of rational powers of the fixed rational number c . By (22), $L = O(\log(m+1))$. The numerators and denominators of all powers $c^{-\ell}$, of $\delta = \eta/m$, and of all grid coordinates have bit length polynomial in $\log(m+1)$. Thus every weight used in the construction has polynomial encoding length.

The representative minimization (31) is solved by a dynamic program. For $\ell \in \{0, \dots, d\}$ and $t \in \{0, \dots, m\}$, store the minimum value of

$$\sum_{r=1}^{\ell} |w_r q_r - g_r|$$

over nonnegative integers $q_1, \dots, q_\ell$ with total count t . Transitioning over the possible value of q_ℓ computes all states in polynomial time. Deterministic lexicographic tie-breaking recovers a unique minimizing count vector. Since (30) gives polynomially many grid points, all representative multisets can be enumerated in polynomial time.

To test retention, write

$$\bar{r}_j = h u_j, \quad u_j \in \mathbb{Z}_{\geq 0},$$

and set

$$K_b = \left\lceil \frac{b}{h} \right\rceil. \quad (59)$$

Because $h = \gamma/m$,

$$K_b = O_{n,\rho}(m).$$

Process the items one at a time. A state is an n -tuple of capped integer loads

$$(\ell_1, \dots, \ell_n) \in \{0, \dots, K_b\}^n.$$

Assigning item j to bin k replaces

$$\ell_k \quad \text{by} \quad \min\{K_b, \ell_k + u_j\}$$

and leaves all other coordinates unchanged. A representative is retained exactly when the terminal state $(K_b, \dots, K_b)$ is reachable. This is equivalent to (39), because an integer load L satisfies $hL \geq b$ exactly when $L \geq \lceil b/h \rceil$. The number of states is

$$(K_b + 1)^n = m^{O(n)},$$

which is polynomial because n is fixed.

10.2 The optimal permutation

Although $\mathcal{F}_m$ contains all assignments of every retained multiset to the goods, these assignments need not be enumerated.

Lemma 5 (Co-sorting). *Fix nonnegative true values v_j , a nonnegative weight multiset $(r_1, \dots, r_m)$, and a threshold τ . Among all assignments of the weights to the goods, the quantity*

$$\min\{v(S) : r(S) \geq \tau\} \quad (60)$$

is maximized by assigning larger weights to goods with larger true values. Ties may be resolved arbitrarily.

Proof. Consider an assignment containing an inversion: for two goods p, q ,

$$v_p \geq v_q \quad \text{but} \quad r_p < r_q.$$

Let r' be obtained by swapping r_p and r_q . We show that the minimum in (60) cannot decrease.

Let S be heavy under r' , meaning $r'(S) \geq \tau$. If S was also heavy under r , use S itself as an old heavy set. Otherwise, the only way the swap can make S newly heavy is that

$$p \in S, \quad q \notin S.$$

Define

$$T = (S \setminus \{p\}) \cup \{q\}.$$

Then

$$r(T) = r(S) - r_p + r_q = r'(S) \geq \tau,$$

while

$$v(T) = v(S) - v_p + v_q \leq v(S).$$

Thus every set heavy under r' has true value at least the old minimum in (60). It follows that the new minimum is no smaller than the old one. Repeatedly removing inversions produces a co-sorted assignment and never decreases the menu floor. $\square$

10.3 Exact share evaluation

For a retained multiset $\bar{r}$, sort its coordinates and the reported item values in the same nonincreasing order, as justified by Lemma 10.1. Write

$$\bar{r}_j = hu_j, \quad K_a = \left\lceil \frac{ab}{h} \right\rceil. \quad (61)$$

Then

$$\bar{r}(S) \geq ab \iff \sum_{j \in S} u_j \geq K_a. \quad (62)$$

Moreover,

$$K_a = O_{n,\rho}(m).$$

The menu floor

$$\min\{v(S) : \bar{r}(S) \geq ab\} \quad (63)$$

is an exact covering-knapsack problem with an $O(m)$ integer threshold. A dynamic program indexed by the processed items and the capped weight

$$\min \left\{ K_a, \sum_{j \in S} u_j \right\}$$

stores the minimum exact rational v -cost achieving each state. Because $\bar{r}$ is retained and $b > ab$, a covering subset always exists. The terminal value at state K_a is exactly (63).

There are polynomially many retained histogram orbits by (30). Evaluating (63) for each orbit and comparing it with the floor $v(\emptyset) = 0$ of F_0 computes (56) exactly in polynomial time. Standard rational addition and comparison remain polynomial because at most m input rationals are summed in any state, so the bit lengths grow only polynomially.

10.4 Allocation

Given a profile $(v_1, \dots, v_n)$, choose for every agent a maximizing menu, using the preceding evaluation algorithm and deterministic tie-breaking. For a selected nonzero menu of agent i , let

$$\bar{r}_{ij} = hu_{ij}$$

be its co-sorted item weights. Every such agent has the common integer threshold K_a from (61). An agent whose maximizing menu is F_0 is assigned threshold zero and is kept inactive during the dynamic program.

Process the goods one at a time. A good may be assigned to one active agent or left unassigned. A state records, for every active agent i , the capped load

$$\ell_i = \min \left\{ K_a, \sum_{j \in A_i} u_{ij} \right\}.$$

There are at most

$$(K_a + 1)^n = m^{O(n)}$$

states. Proposition 7.1 guarantees that a terminal state in which every active coordinate equals K_a exists. Backtracking gives pairwise-disjoint bundles belonging to all selected nonzero menus. Agents selecting F_0 initially receive $\emptyset$. The remaining goods may then be assigned arbitrarily. Every agent's value can only increase, so the completed allocation meets the computed shares.

We have proved the following.

Theorem 3 (Algorithmic lower bound). *For fixed $n \geq 3$ and fixed $\rho < \alpha_n$, the share s_ρ is exactly computable and allocable in time polynomial in the rational input length.*

The algorithm never computes $\text{MMS}_n(v)$. The MMS normalization was used only in the proof that the universal menu collection contains a suitable witness menu.

Combining Theorem 8.2, Theorem 10.2, and the upper bound (6) proves

$$\boxed{\rho_n^* = \alpha_n}. \quad (64)$$

11 A finite exact characterization of α_n

We now show that the answer is itself the value of a finite optimization depending only on n .

Let

$$T = [n]^n.$$

An element

$$t = (t_1, \dots, t_n) \in T$$

will be viewed as an atomic good type. Thus $|T| = n^n$.

Define $\mathcal{X}_n$ to be the set of arrays

$$x = (x_{i,t})_{i \in [n], t \in T} \in [0, 1]^{n \times T}$$

satisfying, for every $i, k \in [n]$,

$$\sum_{\substack{t \in T \\ t_i = k}} x_{i,t} \geq 1, \quad (65)$$

and, for every $i \in [n]$,

$$\max_{\pi: T \rightarrow [n]} \min_{k \in [n]} \sum_{\substack{t \in T \\ \pi(t) = k}} x_{i,t} \leq 1. \quad (66)$$

The maps $\pi : T \rightarrow [n]$ enumerate all ordered partitions of the atomic ground set T , including partitions with empty parts. Hence the left side of (66) is precisely $\text{MMS}_n(x_i)$.

For each fixed agent i , the fibers

$$\{t \in T : t_i = k\}, \quad k \in [n],$$

form a partition of T . By (65), every part of this coordinate partition has value at least 1, so

$$\text{MMS}_n(x_i) \geq 1.$$

Together with (66), this gives

$$\text{MMS}_n(x_i) = 1 \quad (i \in [n]) \quad (67)$$

for every $x \in \mathcal{X}_n$.

The set $\mathcal{X}_n$ is nonempty. For example, let

$$d_k = (k, \dots, k) \in T \quad (k \in [n])$$

and give every agent value 1 for each d_k and value 0 for all other types. Every coordinate fiber contains one unit-valued diagonal type, and the resulting valuation consists of n unit goods, so its MMS is exactly 1.

For $x \in \mathcal{X}_n$, define

$$Q(x) = \max_{\sigma: T \rightarrow [n]} \min_{i \in [n]} \sum_{\substack{t \in T \\ \sigma(t)=i}} x_{i,t}. \quad (68)$$

Here σ ranges over all allocations of the atomic goods.

Theorem 4 (Finite characterization). *For every fixed $n \geq 3$,*

$$\boxed{\alpha_n = \min_{x \in \mathcal{X}_n} \max_{\sigma: T \rightarrow [n]} \min_{i \in [n]} \sum_{\substack{t \in T \\ \sigma(t)=i}} x_{i,t}.} \quad (69)$$

Proof. Let

$$\beta_n = \min_{x \in \mathcal{X}_n} Q(x).$$

The set $\mathcal{X}_n$ is closed: all constraints (65) are closed, and (66) is a closed condition because it involves only finitely many continuous linear functions and finite minima and maxima. It is bounded by the box $[0, 1]^{n \times T}$, and hence compact. The function Q is continuous, again because it is a finite maximum of finite minima of linear functions. Thus the minimum defining β_n is attained.

We first prove

$$\alpha_n \geq \beta_n. \quad (70)$$

Take an arbitrary finite profile $(v_1, \dots, v_n)$ with $\mu_i = \text{MMS}_n(v_i) > 0$ for every i , and normalize it by

$$u_i = \frac{v_i}{\mu_i}.$$

Then

$$\text{MMS}_n(u_i) = 1. \quad (71)$$

For each agent i , choose a partition

$$(P_1^i, \dots, P_n^i) \in \mathcal{P}_n(M)$$

attaining this maximin share, so

$$u_i(P_k^i) \geq 1 \quad (k \in [n]). \quad (72)$$

Assign each original good $j \in M$ the type

$$t(j) = (t_1(j), \dots, t_n(j)) \in [n]^n,$$

where

$$j \in P_{t_i(j)}^i \quad (i \in [n]).$$

Merge all goods of the same type and define

$$U_{i,t} = \sum_{\substack{j \in M \\ t(j)=t}} u_i(j). \quad (73)$$

Now cap each merged value at 1:

$$x_{i,t} = \min\{U_{i,t}, 1\}. \quad (74)$$

We verify that $x \in \mathcal{X}_n$. For every i, k ,

$$\sum_{\substack{t \in T \\ t_i=k}} U_{i,t} = u_i(P_k^i) \geq 1.$$

For any finite collection of nonnegative numbers a_r ,

$$\sum_r \min\{a_r, 1\} \geq \min\left\{\sum_r a_r, 1\right\}.$$

Applying this to the values in the k -th coordinate fiber gives

$$\sum_{\substack{t \in T \\ t_i=k}} x_{i,t} \geq 1,$$

which is (65).

Treating each type as one indivisible atomic good restricts the set of partitions available for u_i : all original goods of a common type must now remain together. Therefore merging cannot increase the maximin share,

$$\text{MMS}_n(U_i) \leq \text{MMS}_n(u_i) = 1. \quad (75)$$

Since $x_{i,t} \leq U_{i,t}$ coordinatewise, capping also cannot increase the maximin share:

$$\text{MMS}_n(x_i) \leq \text{MMS}_n(U_i) \leq 1. \quad (76)$$

The coordinate partition already shows that $\text{MMS}_n(x_i) \geq 1$, so $\text{MMS}_n(x_i) = 1$, and $x \in \mathcal{X}_n$.

Let $\sigma : T \rightarrow [n]$ be any allocation of the atomic instance. Expand it to an allocation of the original instance by assigning every original good of type t to agent $\sigma(t)$. For every agent i , the normalized value in the original instance is

$$\sum_{\substack{t \in T \\ \sigma(t)=i}} U_{i,t} \geq \sum_{\substack{t \in T \\ \sigma(t)=i}} x_{i,t}. \quad (77)$$

Choose σ attaining $Q(x)$. Equation (77) shows that the normalized egalitarian value of the resulting original allocation is at least $Q(x)$, and hence at least β_n . Thus

$$\Phi(v_1, \dots, v_n) \geq \beta_n.$$

The original profile was arbitrary, so taking the infimum proves (70).

For the reverse inequality, every $x \in \mathcal{X}_n$ is itself an additive n -agent instance on the finite ground set T , and (67) says that every agent has MMS exactly 1. Therefore this instance occurs among the profiles in the infimum (1), and

$$\alpha_n \leq Q(x) \quad (x \in \mathcal{X}_n).$$

Taking the minimum over $x \in \mathcal{X}_n$ yields

$$\alpha_n \leq \beta_n. \tag{78}$$

Equations (70) and (78) prove (69). $\square$

12 Rationality and exact computability

We finish by converting (69) into finitely many rational linear programs.

For a fixed agent i , condition (66) is equivalent to

$$\text{for every } \pi : T \rightarrow [n], \quad \text{there exists } k \in [n] \text{ such that } \sum_{\substack{t \in T \\ \pi(t)=k}} x_{i,t} \leq 1. \tag{79}$$

For every pair (i, π) , choose a witnessing bundle

$$k(i, \pi) \in [n].$$

Once all these witnesses are fixed, (79) becomes the finite collection of rational linear inequalities

$$\sum_{\substack{t \in T \\ \pi(t)=k(i, \pi)}} x_{i,t} \leq 1. \tag{80}$$

Enumerating all witness choices expresses $\mathcal{X}_n$ as a finite union of rational polytopes.

Introduce an epigraph variable ζ . By (68),

$$\zeta \geq Q(x)$$

is equivalent to

$$\text{for every } \sigma : T \rightarrow [n], \quad \text{there exists } i \in [n] \text{ such that } \zeta \geq \sum_{\substack{t \in T \\ \sigma(t)=i}} x_{i,t}. \tag{81}$$

For every σ , choose a witnessing agent $i(\sigma)$. With these choices fixed, (81) becomes the rational linear inequalities

$$\zeta \geq \sum_{\substack{t \in T \\ \sigma(t)=i(\sigma)}} x_{i(\sigma),t}. \tag{82}$$

Since $0 \leq x_{i,t} \leq 1$ and $|T| = n^n$, one always has

$$0 \leq Q(x) \leq n^n.$$

We may therefore add

$$0 \leq \zeta \leq n^n. \quad (83)$$

After fixing all witnesses $k(i, \pi)$ and $i(\sigma)$, minimizing ζ subject to (65), (80), (82), (83), and

$$0 \leq x_{i,t} \leq 1 \quad (84)$$

is a rational linear program over a bounded polyhedron. Conversely, the union over all witness choices is exactly the epigraph of Q over $\mathcal{X}_n$; there is no relaxation in this conversion.

For completeness, we record the elementary rational linear-programming fact used here. A nonempty bounded polyhedron described by finitely many rational linear inequalities has a vertex. Every vertex is the unique solution of a full-rank subsystem of active rational linear equalities, and hence has rational coordinates by exact Gaussian elimination. A rational linear objective attains its minimum on a nonempty bounded polyhedron, and its minimum face contains a vertex. Thus the optimum is rational. Moreover, one may solve such a program exactly in principle by enumerating full-rank subsets of the defining inequalities, solving the corresponding rational linear systems, retaining feasible vertices, and comparing their objective values.

There are only finitely many maps

$$\pi, \sigma : T \rightarrow [n]$$

and finitely many possible witness choices. Infeasible witness branches can be discarded, and every feasible branch is a bounded rational linear program of the form just described. Taking the smallest optimum over all branches therefore computes α_n exactly.

Corollary 1. *For every fixed $n \geq 3$, the constant $\alpha_n = \rho_n^*$ is rational, is attained by an atomic profile on at most n^n goods, and is exactly computable by a finite procedure.*

13 Conclusion

The upper bound shows that feasibility alone prohibits every domination factor strictly larger than α_n . The universal menu construction gives, for every $\rho < \alpha_n$, a feasible and self-maximizing share that dominates ρ MMS $_n$, is exactly computable in polynomial time, and admits a polynomial-time allocation algorithm for fixed n and ρ . The finite atomic characterization independently determines α_n by a finite rational optimization.

Consequently, the stated supremum problem is completely solved:

$$\boxed{\rho_n^* = \alpha_n.}$$

The construction does not establish that one admissible share attains α_n itself. This is not a remaining gap in the result, because the quantity requested is the supremum, and every factor strictly below α_n is achieved while every factor strictly above it is impossible.

References

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